SafeMine — Landmine Detection Model
YOLOv8s v2 Report (seed 1)

Generated 2026-10-15 · frozen v2 split · ultralytics 8.4.117

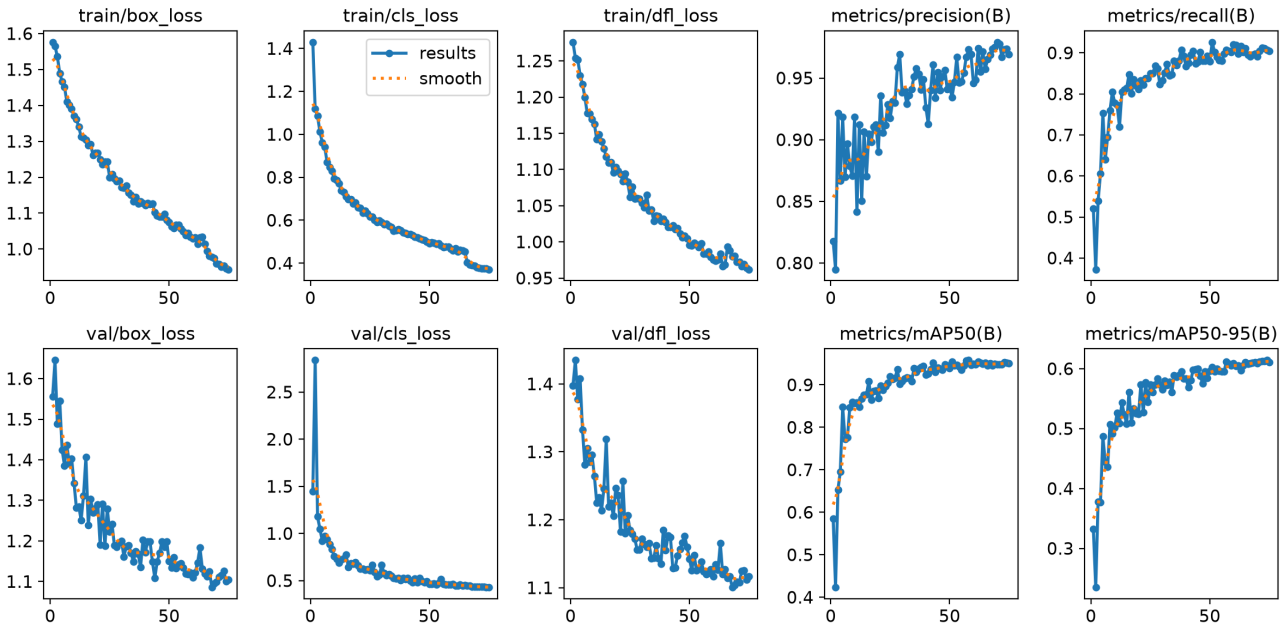
Metric	Validation	Test
Precision (mean)	0.9731	0.9803
Recall (mean)	0.9069	0.9407
mAP@50	0.9523	0.9807
mAP@50-95	0.6134	0.6946
OP conf (val-chosen)	—	0.01
Box P/R @ OP	—	0.3558 / 0.9802
FP/img @ OP	—	2.376
Image-level P/R @ OP	—	0.9839 / 0.9946

Training Configuration

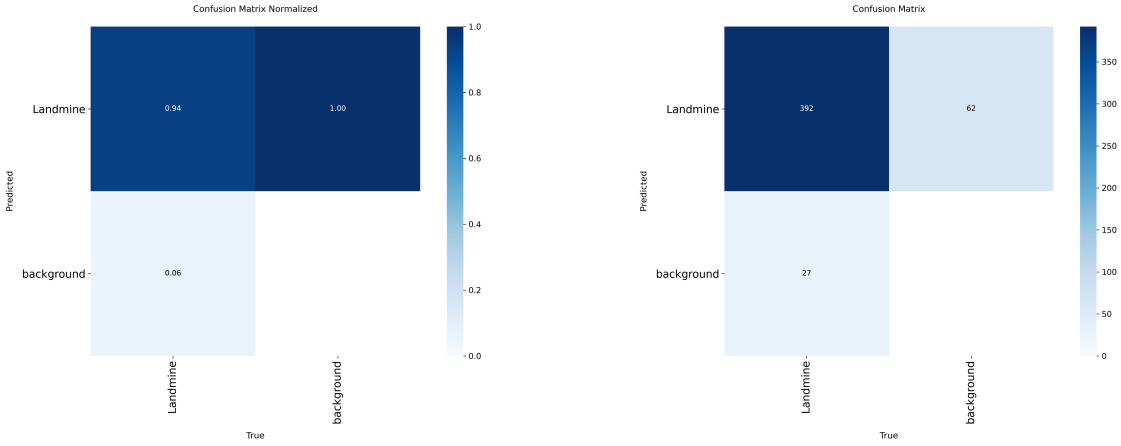
Param	Value
model	yolov8s.pt
epochs	75
imgsz	640
batch	8
seed	1
save_period	5
lr0	0.01
data	C:\Users\Delta\Downloads\Landmine_project\Datasets\landmine_combined_v2\data.yaml

Hardware: NVIDIA GeForce RTX 4070 Laptop GPU (8 GB), FP32, imgsz 640.

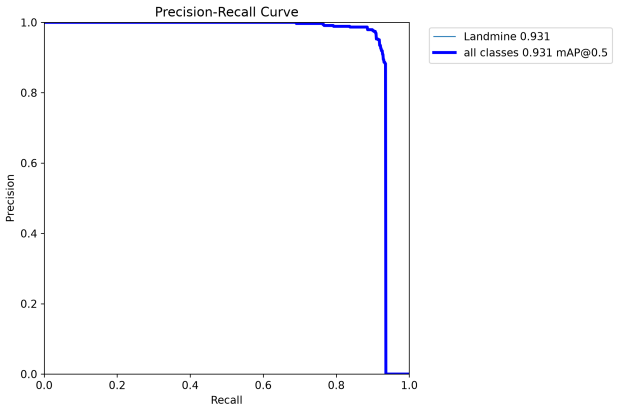
Training Curves

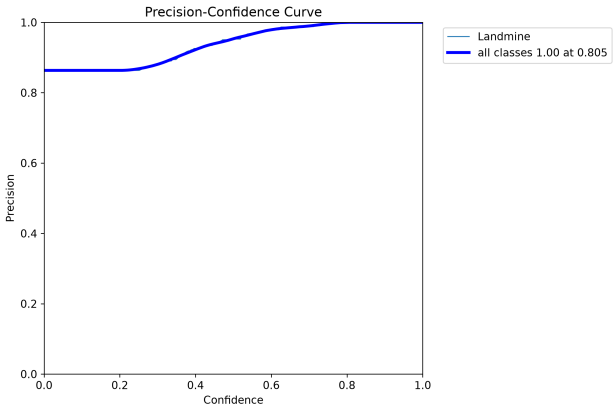


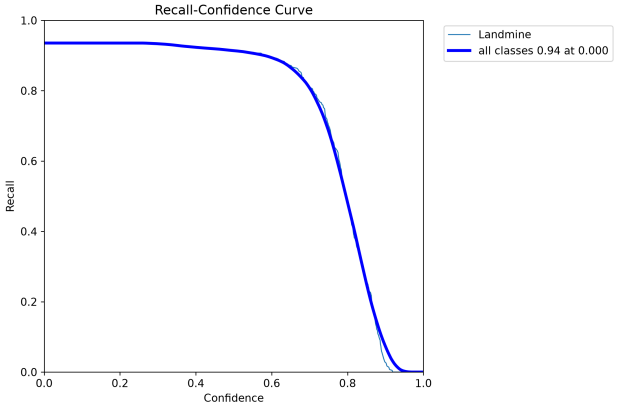
Confusion Matrix (Normalized) — Val

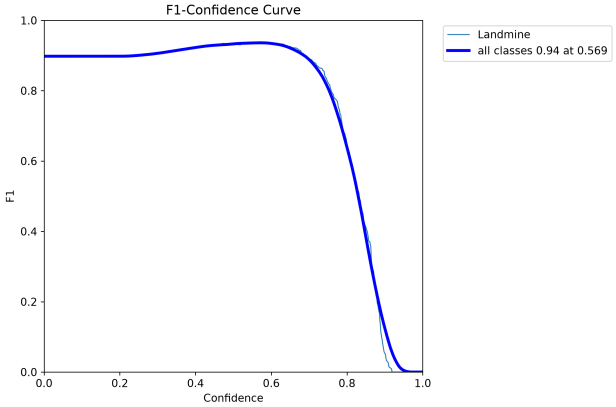


Precision-Recall / P / R / F1 Curves — Val

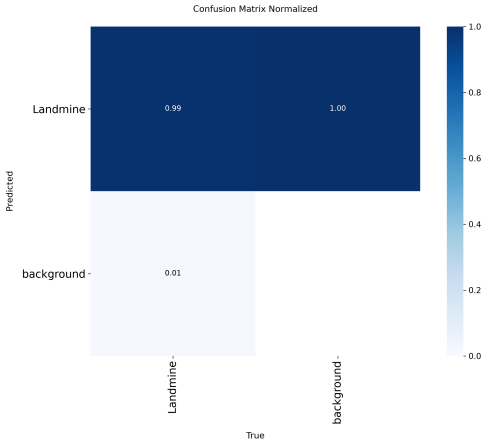


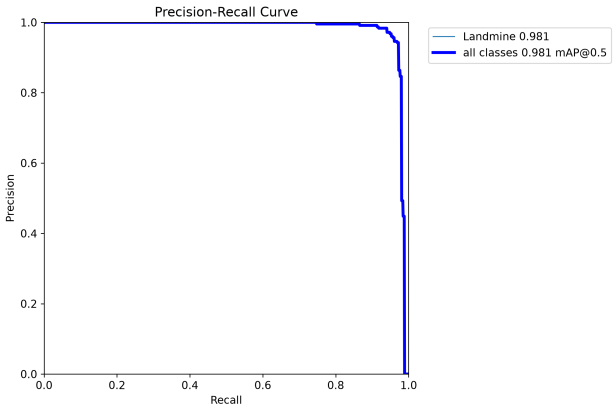




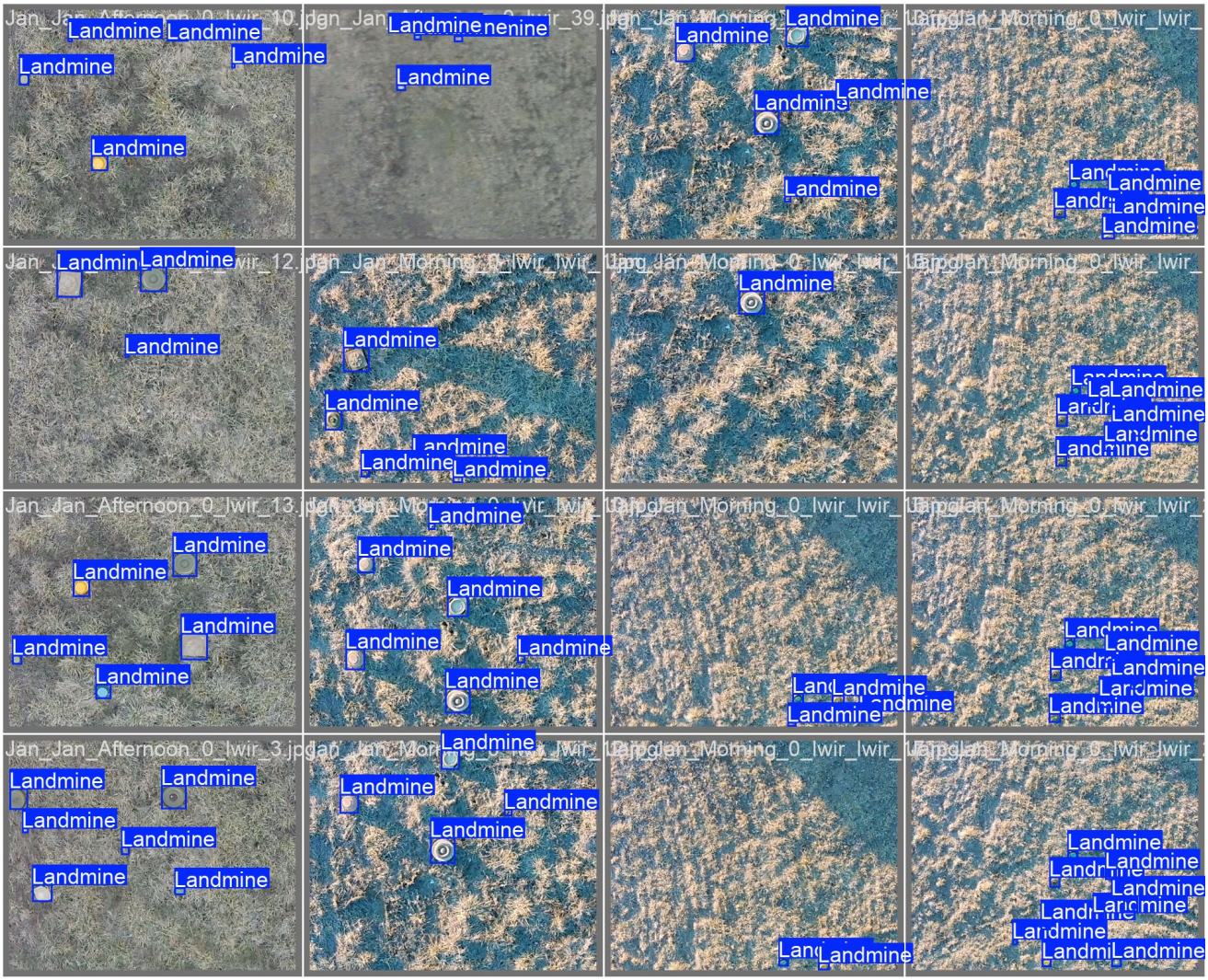


Confusion + Curves — Test (visualization pass @ frozen OP conf)

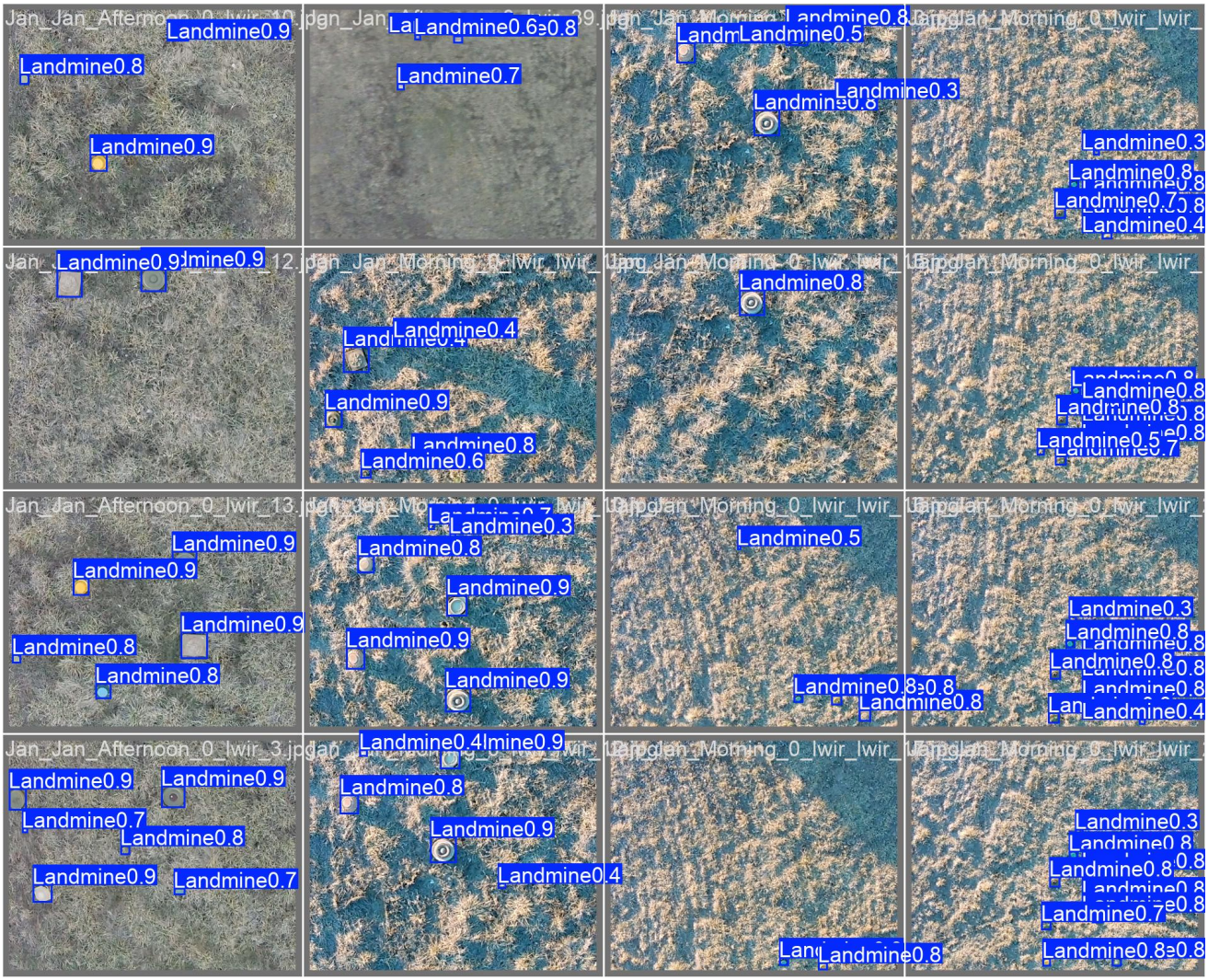




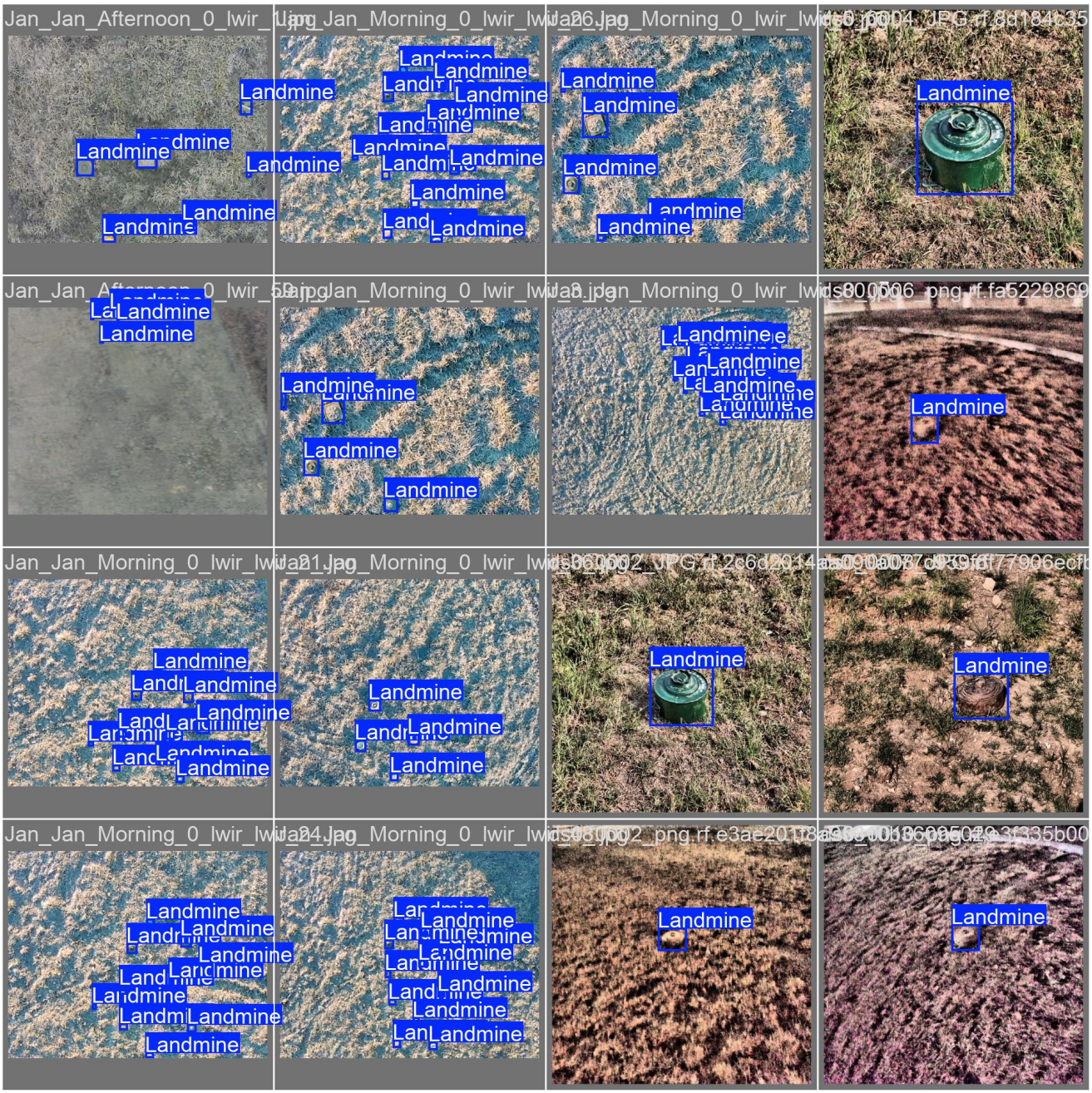
Validation Ground Truth (batch 0)



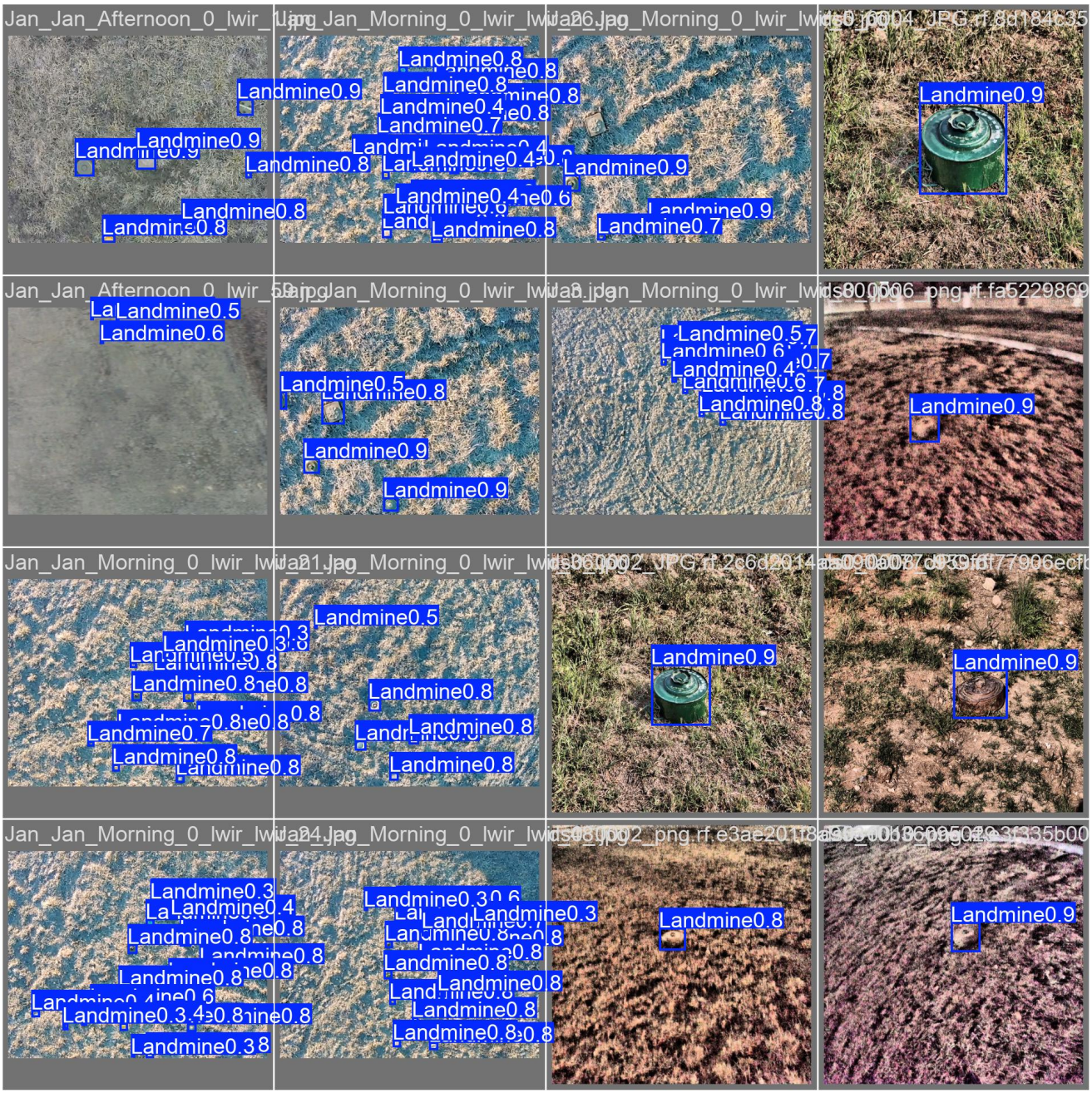
Validation Predictions (batch 0)



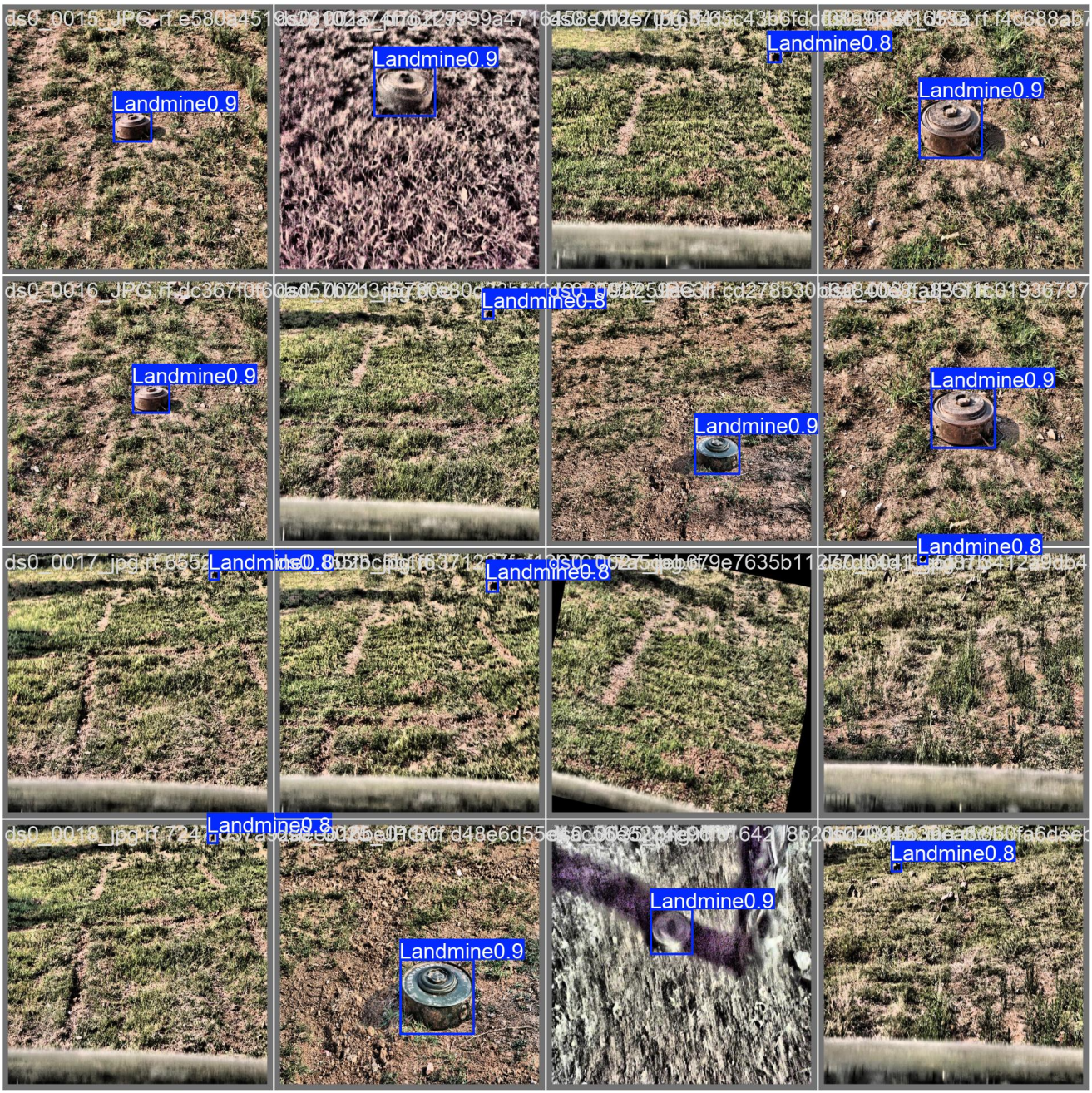
Test Ground Truth (batch 0) — seed 1



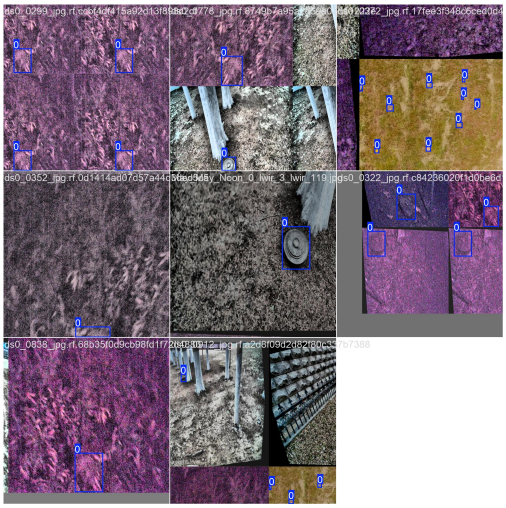
Test Predictions (batch 0) @ conf 0.01 — seed 1

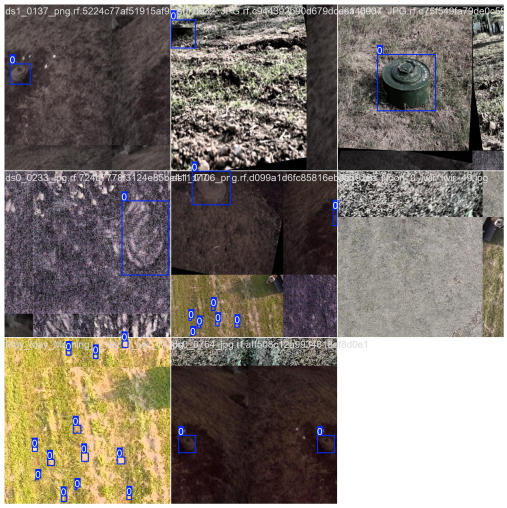


Test Predictions (batch 1) @ conf 0.01 — seed 1



Training Augmentation Sample





Official metrics come from paper_results/yolo/s_seed1/{val,test}.json (frozen protocol; OP conf chosen on val only). Test batch images above are from a visualization-only pass (make_test_vis.py); they do not change any reported number.